

High Injury Network Analysis

Avalon Borough, Cape May County, New Jersey

Analysis Period: 2017 - 2021

Report Generated: September 10, 2026

Executive Summary

This report presents the results of a High Injury Network (HIN) analysis for Avalon Borough, conducted for the period 2017 through 2021. The analysis screens roadway segments for elevated recorded crash counts per mile-year. Severity scores are separate descriptive rankings. Candidate locations require professional review before safety investment decisions.

Key Findings

- Total crashes analyzed: 133
- Fatal crashes: 0
- Serious injury crashes (classified records only): 0
- Total fatalities: 0
- Total injuries: 41
- High Injury Network length: 0.9 miles
- HIN segments identified: 9

Crash Statistics

Severity	Count	Percentage
Fatal	0	0.0%
Serious Injury	0	0.0%
Minor Injury	0	0.0%
Injury - detail unknown	37	27.8%
Property Damage	96	72.2%
Total	133	100.0%

Vulnerable Road Users

Category	Killed	Injured
Pedestrians	0	5
Bicyclists	Unknown	Unknown

Unknown means the source does not provide complete counts. Injury severity is not split into serious/minor in NJDOT Accidents archives. Bicycle casualties cannot be inferred from involvement flags. Loaded records exclude crashes without a usable location; these are not complete crash totals.

High Injury Network Results

These candidate segments have elevated counts of all recorded crashes relative to a municipality-local reference rate. Severity scores are descriptive rankings, not Poisson event counts. This exploratory screening is not an independently validated or grant-certified High Injury Network.

Top 10 High Injury Network Segments

Rank	Road Name	Crashes	Rate	Length (mi)
1	05000619__	15	30.49	0.10
2	05000619__	4	8.13	0.10
3	05000619__	4	8.13	0.10
4	05011001__	3	6.43	0.09
5	05000601__	3	6.21	0.10
6	05011075__	3	6.13	0.10
7	05011075__	3	6.13	0.10
8	05011075__	3	6.13	0.10
9	05011098__	3	6.05	0.10

Methodology

This exploratory screening uses loaded, locatable crash records. It is not independently validated for grant eligibility. Traffic exposure, spatial dependence, over-dispersion, and multiple testing are not modeled; professional review is required.

Analysis Steps

- Crash Data Collection: Official crash records from NJ Department of Transportation
- Spatial Processing: Crashes snapped to the nearest road segment within 50 meters
- Severity Weighting: Descriptive ranking only; fatal: 10, serious injury: 5, minor injury: 3, property damage: 1; unspecified injury detail: 3
- Rate Calculation: Crash rates calculated per segment mile per year
- Statistical Testing: One-sided Poisson screening on actual crash counts, minimum 3 crashes. Leave-one-out road-class baseline, then municipality fallback; no reference exposure means no selection.
- Significance Threshold: P-value < 0.05
- Network Assembly: Same-route endpoints within 1 meter form connected corridors. Fragments shorter than 0.01 mile are excluded from screening and baseline exposure, not deleted.
- Social vulnerability: Official CDC/ATSDR SVI 2020 national tract rankings where loaded; a contextual snapshot, not a measure for each crash year. Missing scores remain unknown.

Data Sources

- Crash Data: NJ Department of Transportation (NJDOT) county/year Accidents archives
- Road Network: NJDOT measured roadway network; route-milepost locations are estimates using that network vintage
- Municipal Boundaries: NJ Office of GIS